

Introduction to numerical Methods - 8 -

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Fitting

Generally minimize Square error

Data: x_i, y_i

Fitted function $f_{\theta}(x)$

θ : list of params

e.g. $f(x) = ax + b$

$\theta = \{a, b\}$

$$S(\theta) = \sum_i (y_i - f_{\theta}(x_i))^2$$

$$\theta_{fit} = \underset{\theta}{\text{argmin}} S(\theta)$$

If data includes σ_i uncertainty of data point i

Sum of squared residuals

$$S(\theta) = \sum_i \left(\frac{y_i - f_{\theta}(x_i)}{\sigma_i} \right)^2$$

Coefficient of determination, R^2

$$SS_{res} = \sum_i (y_i - f_{\theta}(x_i))^2$$

$$\bar{y} = \frac{1}{n} \sum_{i=1}^n y_i$$

$$SS_{tot} = \sum_i (y_i - \bar{y})^2$$

$$R^2 = 1 - \frac{SS_{res}}{SS_{tot}}$$

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Problems: if y values are very different in scale

- Exponential function: solution: semi log
 $y = ae^{\beta x}$ $\log y = \log a + \beta x$

- Power law: solution: log-log
 $y = ax^b$ $\log y = b \log x + \log a$

Error Propagation

- Sources of error in data

- Precision of instrument
- Human error
- Stochastic nature of phenomenon
- Environment

- Error types

- Random: repeating experiment reduces random errors
- Systematic: change environment, vary parameters to detect errors, but nonetheless they stay

Random error

$$\bar{x} = \frac{1}{n} \sum_{i=1}^n x_i$$

$$s = \sqrt{\frac{1}{n} \sum_{i=1}^n (x_i - \bar{x})^2}$$

Sample standard deviation

$$s_c = \sqrt{\frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})^2}$$

(n-1 includes a downward correction (-1) to correct the downward bias of the Concave square root)

Standard deviation of the mean

$$s_{\bar{x}} = \sqrt{\frac{1}{n(n-1)} \sum_{i=1}^n (x_i - \bar{x})^2}$$

Thus the std of the mean is reduced by $\sqrt{n}$

- 68% confidence interval : $\bar{x} \pm 1s_i$
- 95% ————— " ————— : $\bar{x} \pm 2s_i$
- 99.7% ————— " ————— : $\bar{x} \pm 3s_i$
- 99.9999998% ————— " ————— : $\bar{x} \pm 6s_i$

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Two variable operations: $f(a, b)$

$$f \approx f^0 + \frac{\partial f}{\partial a} a + \frac{\partial f}{\partial b} b + \dots$$

$$\text{Var}(aX + bY) = a^2 \text{Var}(X) + b^2 \text{Var}(Y) + 2ab \text{Cov}(X, Y)$$

$$\sigma_f^2 \approx \left(\frac{\partial f}{\partial a} \right)^2 \sigma_a^2 + \left(\frac{\partial f}{\partial b} \right)^2 \sigma_b^2 + 2 \frac{\partial f}{\partial a} \frac{\partial f}{\partial b} \sigma_{ab}$$

if a and b are uncorrelated, then $\sigma_{ab} = 0$

For independent case

- Multiplication by constant:

$$f(a) = Ca, \quad \sigma_f = |C| \sigma_a$$

- Addition: $f(a, b) = a \pm b$ derivatives are ± 1

$$\sigma_f = \sqrt{\sigma_a^2 + \sigma_b^2}$$

This translates to the $\frac{1}{\sqrt{2}}$ for the

$$\text{average: } f(a, a) = \frac{a+a}{2} \Rightarrow \sigma_f = \frac{1}{2} \sqrt{2\sigma_a^2} = \frac{1}{\sqrt{2}} \sigma_a$$

- Product: $f(a, b) = a \cdot b$

$$\sigma_f = |f| \sqrt{\left(\frac{\sigma_a}{a} \right)^2 + \left(\frac{\sigma_b}{b} \right)^2}$$

- Power: $f(a) = a^t : \sigma_f = |f| \sigma_a / a$

Root finding

$$- f(x) = 0$$

- under:

$$f(x_1, \dots, x_n) = 0$$

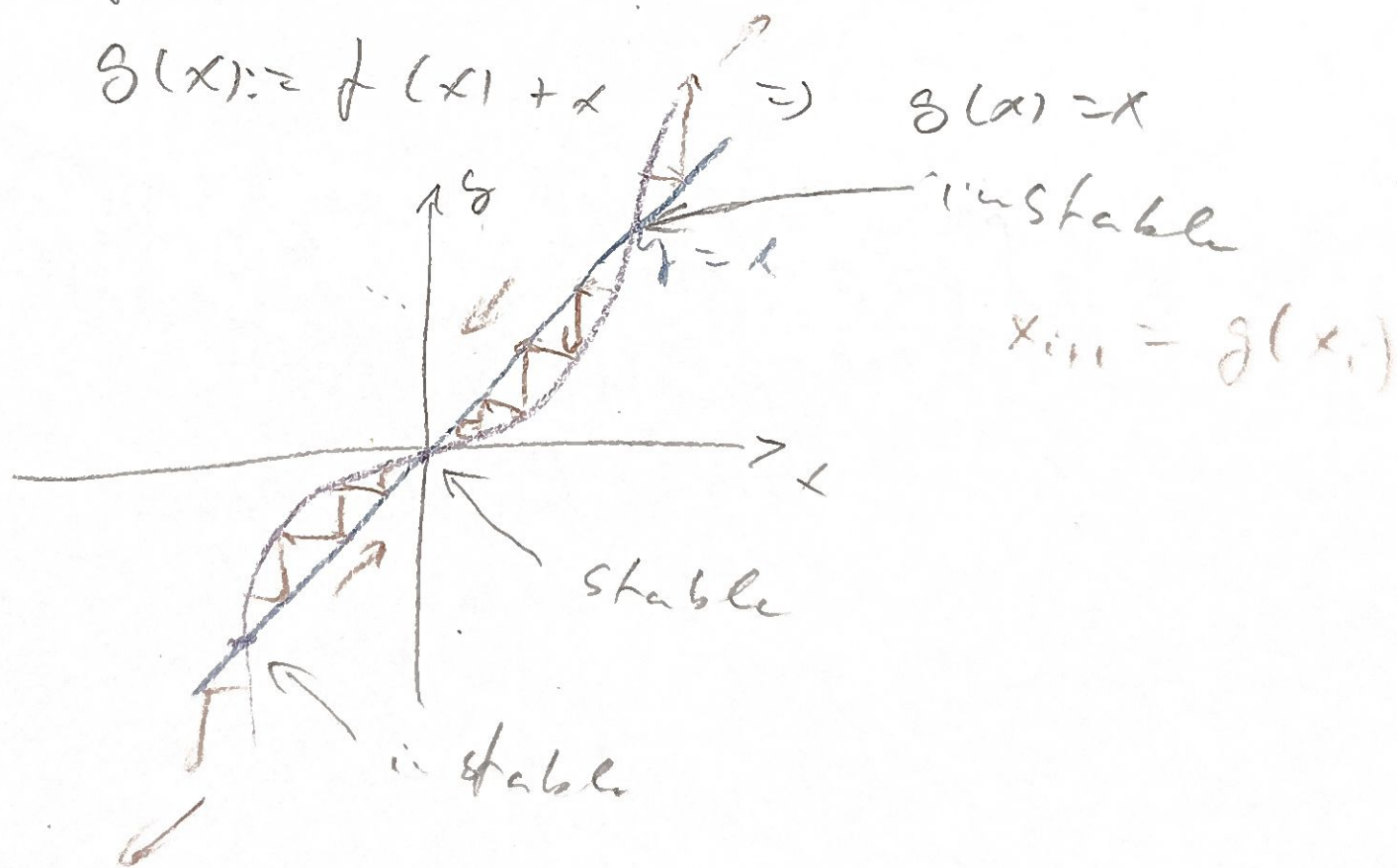
$$f_1(x_1, \dots, x_n) = 0$$

$$f_n(x_1, \dots, x_n) = 0$$

Fixed point method

$$f(x) = 0$$

$$g(x) := f(x) + x \Rightarrow g(x) = x$$



Two types of fixed points: - stable
- unstable

In unstable fixed points the derivative of f is negative
Solution $f' = -1$ to get the unstable ones

Stop at required precision

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Starting point: good question;

Try to map areas of convergence

Bisection method

Idea is for (x_1, x_2) such that $\text{sgn}(f(x_1)) \cdot \text{sgn}(f(x_2)) < 0$, then there must be at least one root on the interval (x_1, x_2)

Method: bisect the interval

$$x_0 = \frac{x_1 + x_2}{2}$$

Assign the value of x_0 to x_1 or x_2 which of the two has the same sign as x_0 .

$$x_0 \rightarrow \begin{cases} x_1 & \text{if } \text{sgn}(x_1) = \text{sgn}(x_0) \\ x_2 & \text{otherwise} \end{cases}$$

We stop at required precision